

Brush-border membrane cation conducting channels from rat kidney proximal tubules

SHIMON MAROM, DANIEL DAGAN, JOSEPH WINAVER, AND YORAM PALTÍ

Department of Physiology and Biophysics, Faculty of Medicine,

and the Rappaport Institute for Research in the Medical Sciences, Technion, Haifa 31096, Israel

MAROM, SHIMON, DANIEL DAGAN, JOSEPH WINAVER, AND YORAM PALTÍ. *Brush-border membrane cation conducting channels from rat kidney proximal tubules.* *Am. J. Physiol.* 257 (Renal Fluid Electrolyte Physiol. 26): F328–F335, 1989.—This is a description and kinetic characterization of cation channels from rat kidney brush-border membrane vesicles and from apical membranes of proximal tubule cells in culture. Channel activity was demonstrated and characterized in both artificial phospholipid bilayers and in tissue culture. Intermediate conductance, ~50 pS, cation-selective channels were observed by both methods. Channels were characterized by a Na permeability (P_{Na})/K permeability (P_K) of 1–5:1. Open-channel current-voltage curves were linear in symmetric 300 mM NaCl. In tissue culture the gating kinetics are described by two open-time constants and two closed-time constants. Channel activity was neither voltage nor Ca^{2+} dependent and the probability of being in the open state ranged from 0.6 to 0.95. In tissue culture experiments the channel demonstrated nonstationary gating activity. A second, 15-pS cation channel, seen in planar bilayers, demonstrated a higher selectivity for Na^+ with a (P_{Na}/P_K ratio of >10).

patch clamp; membrane vesicles; primary culture; model membrane; ion channels

THE APICAL MEMBRANE of epithelial cells from mammalian renal proximal convoluted tubules (PCT), are inaccessible to direct measurements of ionic transport *in situ*. Several preparations, including brush-border membrane vesicles (BBMV) and epithelial cell cultures are being used to circumvent this problem. The use of these preparations has been limited primarily to studies at the level of macroscopic ion fluxes. In an attempt to gain further understanding of transport mechanisms in these membranes, we resorted to two single-channel monitoring techniques; reconstitution of single channels from BBMV into artificial lipid membranes and patch-clamping of apical membranes of PCT epithelial cells in culture. Two complementary variants of lipid bilayer systems, one in a chamber partition and the other at the tip of a patch electrode, “tip dipping,” were employed for channel reconstitution.

A reconstituted channel in a bilayer is a well-defined system since the channel is infinitely diluted and isolated from all other cellular components. Cell cultures on the other hand provide a model system approximating more closely native conditions. Both of these approaches provide access to the apical membrane channels and serve as a means for ionic current analysis at the molecular

level. Characteristics including conductance, selectivity, and gating kinetics of single channels from the inaccessible apical membranes *in situ* can thus be obtained and compared under the two experimental conditions. We report here that a variety of cation-permeable channels derived from apical membranes of rat PCT epithelial cells can be reconstituted in both “tip dip” and conventional planar bilayers. Epithelial cells from the same source grown in culture, demonstrated cation channel activity in both cell-attached and inside-out patches of apical membranes.

METHODS

Preparation of BBMV. Male Sprague-Dawley rats (250–300 g body wt) were anesthetized with pentobarbital sodium (35 mg/kg). Their kidneys were removed and placed in ice-cold mannitol buffer [10 mM mannitol and 2 mM tris(hydroxymethyl)aminomethane (Tris)·HCl, pH 7.1]. After decapsulation thin slices of renal cortex were pooled from four to six animals. BBMV (Fig. 1A) were isolated by calcium or magnesium precipitation methods (1, 7). Protein concentration was determined by a modified Lowry microassay (11). For experiments carried out in model membranes at the tip of patch-clamp electrodes, the native membrane vesicles were resuspended in 200 mM NaCl and 25 mM *N*,2-hydroxyethyl piperazine-*N*,*N*-tetraacetic acid (HEPES), pH 7.4, and mixed with a sonicated suspension of bovine brain phosphatidylserine (PS) and phosphatidylethanolamine (PE) (Avanti Polar Lipids, Birmingham, AL) in a 5:2 wt/wt mixture to give a lipid/protein ratio of 10–50/1. The suspension was freeze-thawed three times in the presence of 0.1 mM $CaCl_2$. The final solution contained 4–20 mg/ml lipids, 0.4 mg/ml protein, 200 mM NaCl, 0.1 mM Ca^{2+} as $CaCl_2$, and 25 mM HEPES, pH 7.4. For experiments carried out in planar lipid bilayer chambers, native membrane vesicles were suspended in 800 mM glucose and 20 mM HEPES, pH 7.4, to give 3–5 mg protein/ml.

Formation of lipid bilayers in a chamber partition. For detailed description see Ref. 4. A mixture of PE/PS (1:1; wt/wt) in decane (20 mg/ml) was used as a stock solution. The membrane was painted by a club-shaped glass rod over a 300- μ m hole in a polystyrene partition separating two compartments. The *cis* solution contained 300 mM NaCl, whereas the *trans* solution contained 150 mM NaCl, creating an osmotic gradient. Both solutions were buffered by 20 mM HEPES to pH 7.4. After allowing the

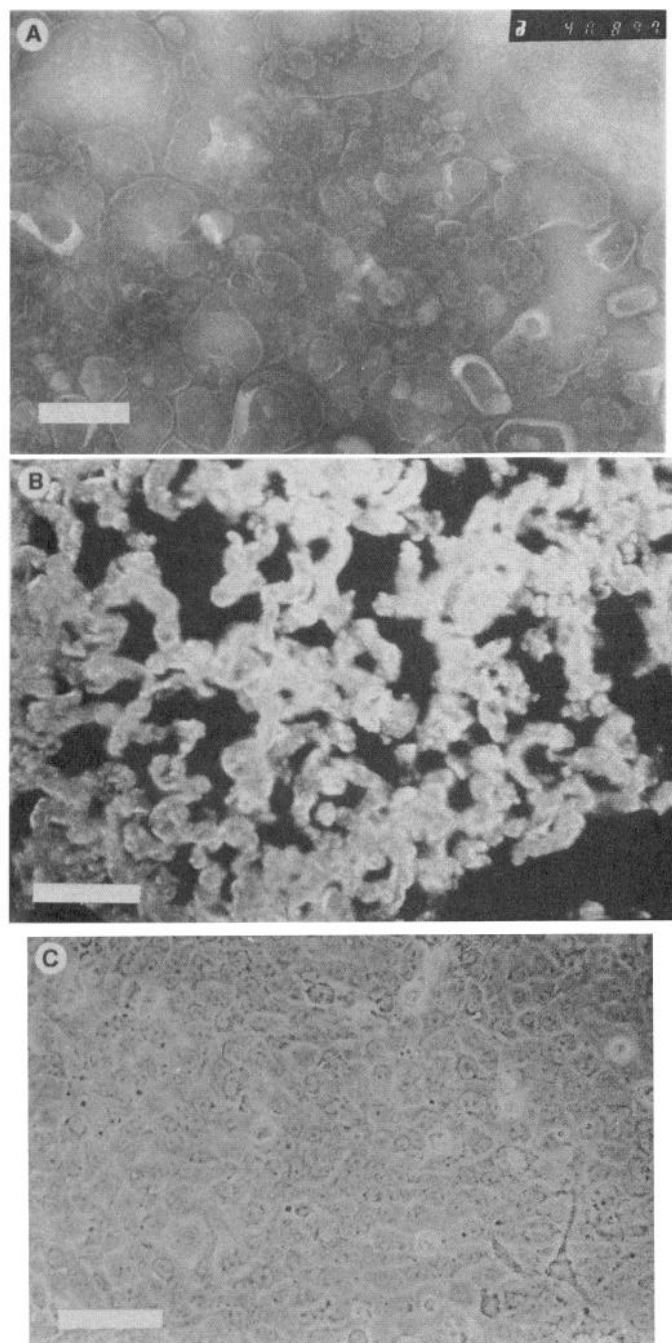


FIG. 1. A: negative staining of brush-border membrane vesicles (scale bar, 0.5 μ M). B: phase micrograph of proximal convoluted tubule fraction purified in Percoll gradient (scale bar, 200 μ m). C: epithelial monolayer 4 days in culture (scale bar, 40 μ m).

bilayer to thin, monitored by its capacitance to >300 pF, we tested the membrane for absence of channel-like activity. When this criterion was satisfied, 20 μ l of native membrane preparation was added to the *cis* side while the chamber was thoroughly stirred. In most cases Ca^{2+} was added to a final concentration of 3 mM to enhance fusion. Fusion usually occurred within 30 min. When channel gating was observed, stirring was stopped and ethylene glycol-bis(β -aminoethyl ether)-*N,N,N',N'*-tetraacetic acid (EGTA) was added to chelate the Ca^{2+} . The experimental solutions were then added to the *cis* side.

TABLE 1. Solutions used in tissue culture experiments designed to distinguish channels according to selectivity

Ion	Pipette, mM	Medium, mM	Cell, mM	On Cell, E_{rev} , mV	Detached, E_{rev} , mV
Na	10	150	10	0	-68
K	70	5	70	0	66
Cl	80	157	10	-52	17
Ca	0*	1	10^{-7}		

Glucose (150 mM) was added to balance osmolarity. See Refs. 10 and 14 for detailed information on cell experiments. On cell values are absolute potentials. Resting potentials should be subtracted to compute pipette potentials. Note that for an ideal nonselective cation channel the reversal potential (E_{rev}) is 0 mV (absolute potential) in the on-cell mode and the cell-detached mode, whereas for an ideal anion channel the reversal potential is -52 mV (absolute potential) in the on-cell mode and +17 mV (absolute potential) in the detached mode. * 5 mM EGTA was added.

Formation of lipid bilayers at tips of patch pipettes. For a detailed description see Refs. 2 and 3. Vesicle suspension (0.25 ml) was placed in one 0.30-ml well (*cis* side) of a sterile microtiter plate (M29A4, Sterilin, Middlesex, UK). Five to ten minutes were allowed for vesicles to form a monolayer at the water-air interface. The same monolayer was used repeatedly for up to 2 h to form membrane bilayers. Electrodes, 5–10 M Ω , were pulled and fire polished from borosilicate glass capillaries (Jencons H15/10). The pipette was passed twice through the monolayer to form a bilayer at its tip. Once channel activity was observed, the pipette was transferred to an adjacent well containing the experimental solution. Seal resistances were usually >20 G Ω and stable for up to 20 min. The same pipette was used several times for formation of bilayers. Potential differences in the range of 150 mV (well relative to pipette) were applied across the bilayer. Higher voltages usually resulted in disruption of the seal. As a control, a pure liposome suspension was used to test for absence of channel-like activity.

Preparation of rat PCT cell cultures. To isolate a pure fraction of rat PCT (Fig. 1B), the procedure described in Ref. 13 was used with some modifications. Two male Wistar (200–250 g) rats were used for each preparation. Their kidneys were removed and decapsulated under sterile conditions, cut in half, and the medulla was dissected out. With the use of a razor blade, the cortex was sliced and treated with a 15-ml dissection solution (DS) supplemented with 0.1% collagenase (type I, Sigma) and 5 mg bovine serum albumin (BSA) (Sigma). (DS contained 140 mM NaCl, 3 mM KCl, 1 mM NaH_2PO_4 , 1 mM MgCl_2 , 2 mM CaCl_2 , 5.5 mM glucose; pH 7.4 titrated with NaOH). The enzymatic treatment was carried out in a 40-rpm shaking bath for 60 min at 37°C. Digestion was stopped by addition of 20 ml cold DS and the suspension was filtered through a tea strainer. The tissue suspension was then centrifuged at 60 *g* for 30 s. The supernatant was discarded and the tissue was resuspended in cold DS and centrifuged again. This procedure was repeated four times. BSA (5%) was added to the DS in the last wash. The tissue was then resuspended in 50% Percoll-DS that was gassed for 30 min with 95% O_2 -5% CO_2 and spun for 30 min at 12,000 *g*, 4°C (Sorvall SS-3 rotor). The tissue separated into four bands. The

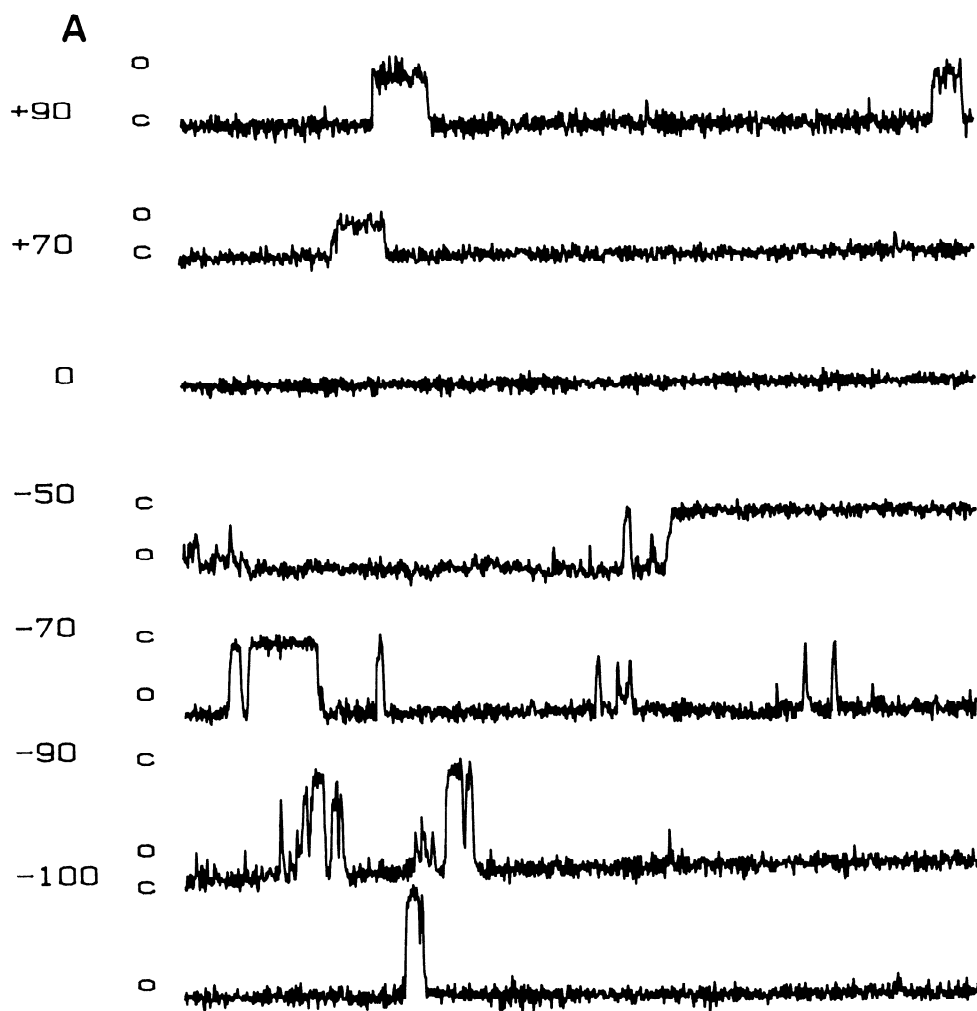
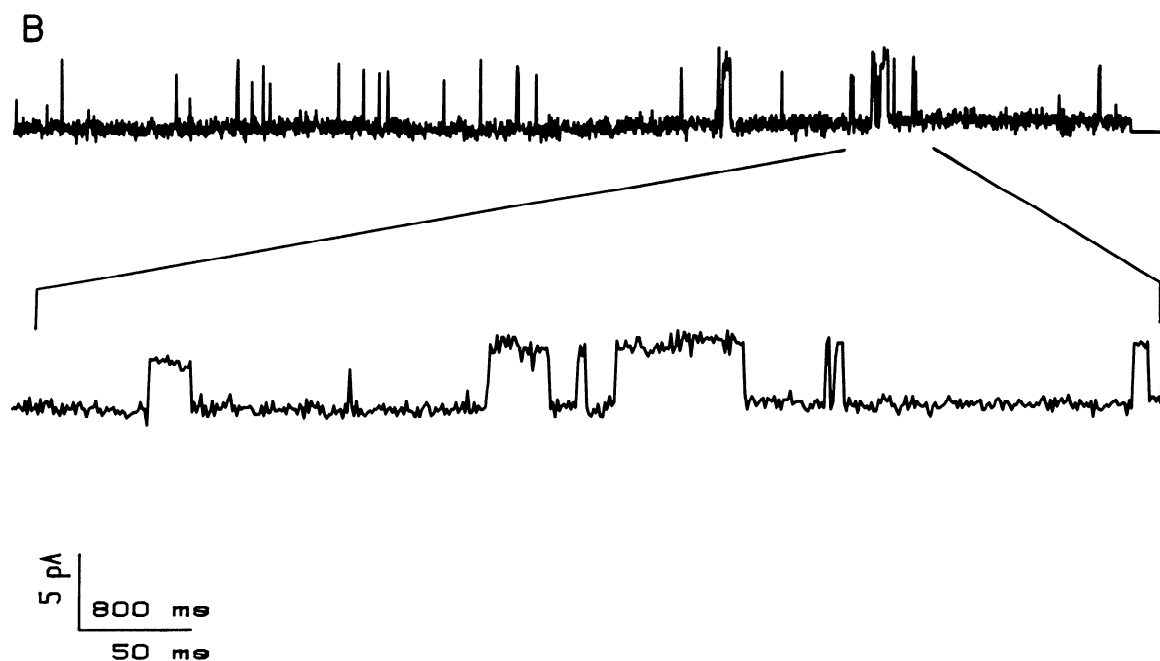


FIG. 2. Brush-border membrane vesicle cation channel activity in planar bilayers. A: 60-pS channel. *Cis* solution contained 300 mM NaCl, 1 mM CaCl_2 , and 20 mM HEPES (pH 7.4). *Trans* solution contained 150 mM NaCl, 0.5 mM CaCl_2 , and 10 mM HEPES (pH 7.4). Voltages on left are given as *trans* relative to *cis*. Downward current stands for movement of cations from *cis* to *trans*. Scale, 4 pA, 100 ms. B: gating pattern of 60-pS channel demonstrating bursting behavior. A selected section is shown on an expanded time scale (bottom trace). Upward deflections denote channel openings. Recorded under symmetrical 140 mM KCl. $V = 50$ mV. C: 15 pS channel. Solutions and conventions as in A. Top trace, $V = +40$ mV; bottom trace, $V = -40$ mV. Scale, 0.5 pA, 200 ms.



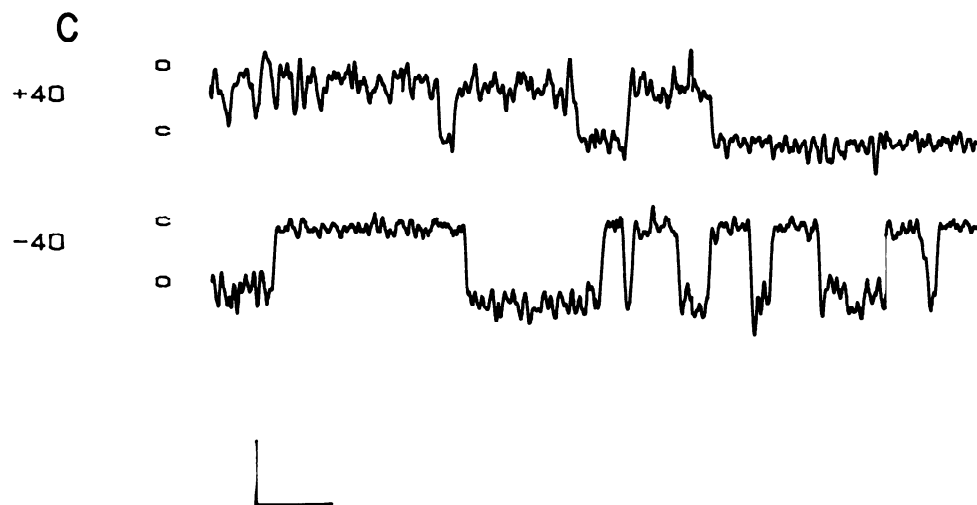


FIG. 2. See facing page for legend.

bottom band contained the PCT fragments. This band was resuspended in 1–2 ml of Dulbecco's modified Eagle's medium (DMEM) (Bet-Haemek, Israel). Twenty microliters of the suspension were then added to each of 20 tissue culture dishes (35 mm, no. 153066, Nunclon, Denmark) containing 2 ml DMEM supplemented with 10% FCS (Bet-Haemek) and 5% antibiotic antimycotic mixture (10,000 units/ml penicillin, 10 mg/ml streptomycin, and 250 μ g/ml amphotericin B, Bet-Haemek). Culture

plates were not coated with collagen or any other substrate, nor were any cells used as an underlayer. Several experiments were conducted without the addition of the antibiotic antimycotic mixture to exclude amphotericin-induced channels. Similar results were obtained under these conditions. All supplements were added to the medium immediately before use. The tissue culture was kept at 37°C in 5% CO₂. Monolayers containing up to 100 cells were present after 48 h (Fig. 1C). The experi-

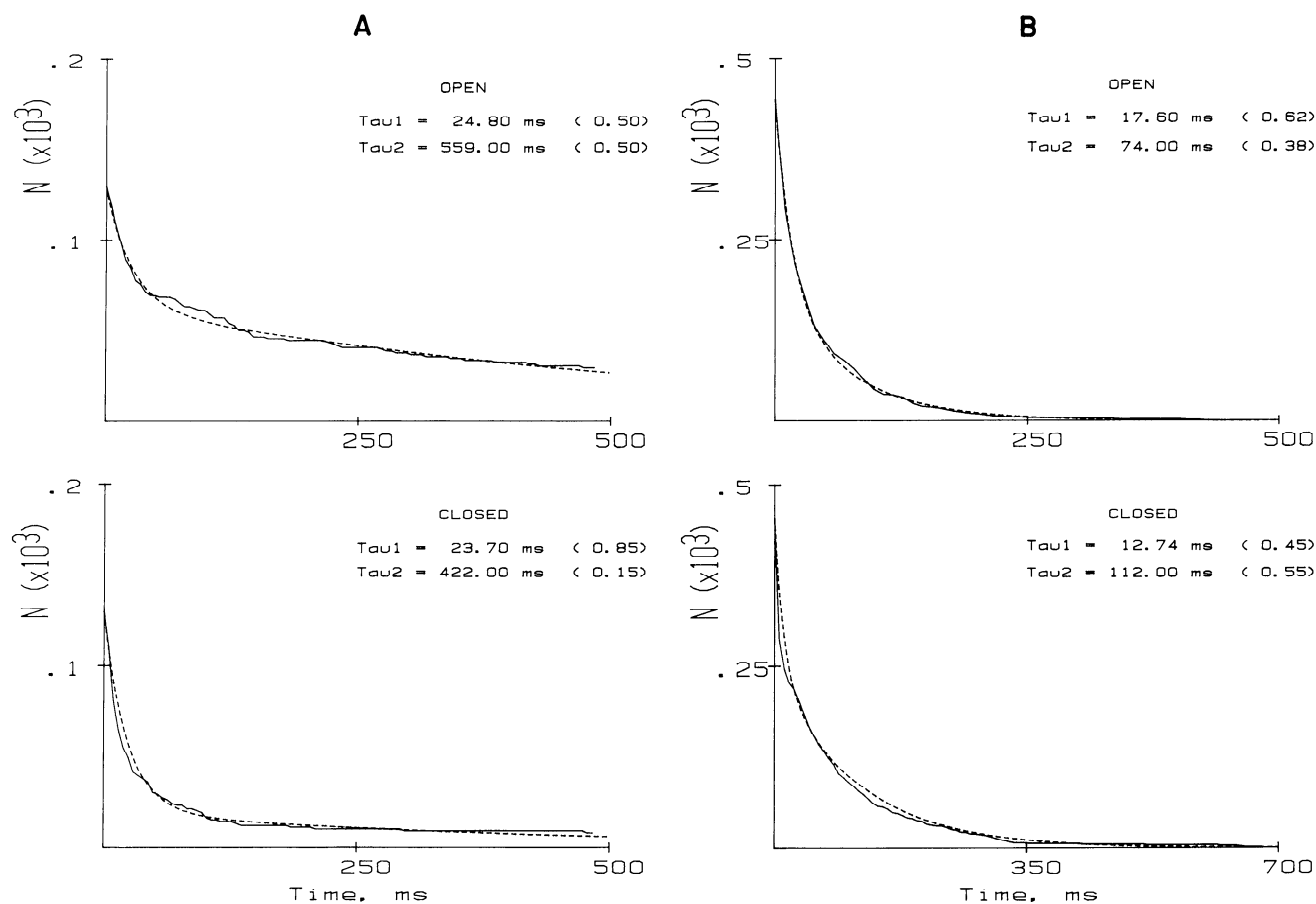


FIG. 3. A: open- and closed-time distributions for 60-pS channel. Bin size = 5 ms, $V = -50$ mV. B: open- and closed-time distributions for 15-pS channel. Bin size = 7 ms, $V = -40$ mV; filtered at 300 Hz. τ_{01} and τ_{02} , time constants. Values in parentheses represent relative coefficient of each exponent in overall activity.

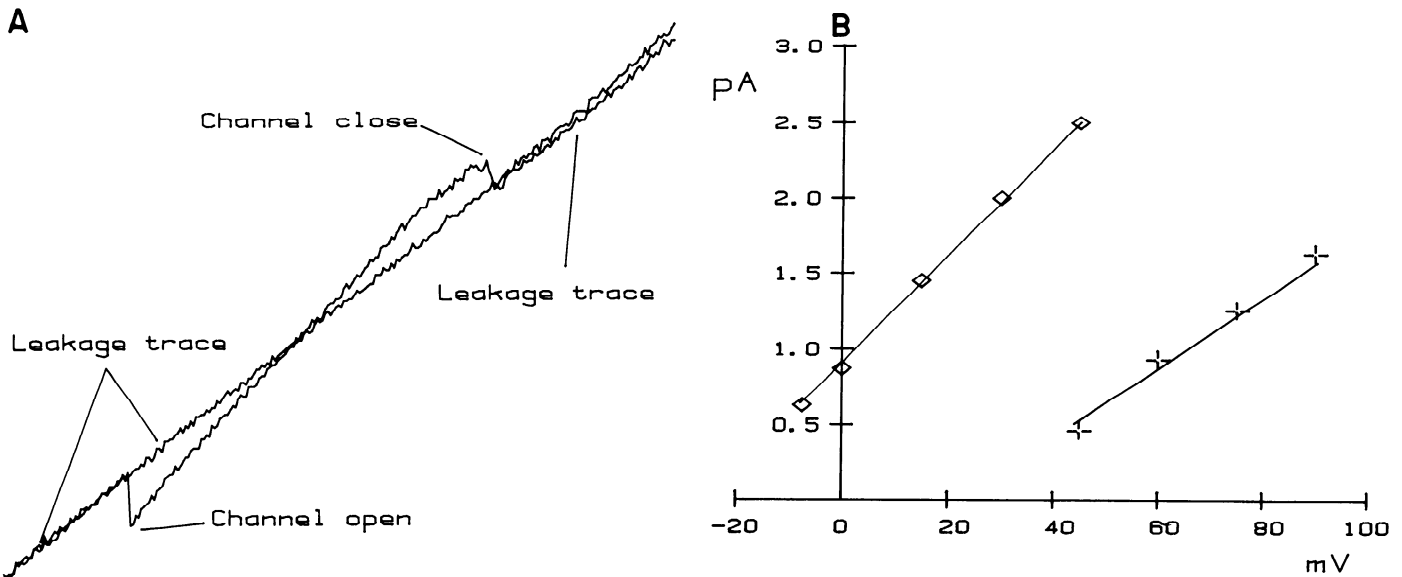
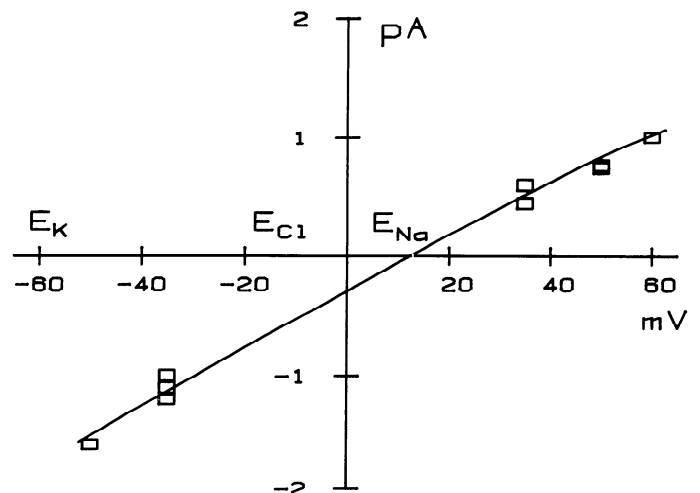


FIG. 4. A: currents induced by clamping membrane with a voltage ramp from -80 to 80 mV, recorded in a bilayer on tip of a patch-clamp pipette. Top part shows raw data of 2 traces. In 1 trace, no channel opening events are seen. Other trace includes 1 opening and 1 closing event. Trace without gating activity is defined as "leakage trace." This trace is subtracted from trace containing an opening event. Leak-corrected trace is put on I - V axes to compute conductance and reversal potential (bottom). All steps are computer controlled. In record shown, pipette and well solutions were 140 mM KCl, 5 mM NaCl, and 140 mM NaCl, 5 mM KCl, respectively. Reversal potential of -7 mV yields a $P_K/P_{Na} = 0.7$ computed using Goldman-Hodgkin-Katz equation. In voltage range of -48 to 20 mV conductance is linear (50 pS). B: I - V relationship of channel under different ionic conditions. For right hand I - V curve (crosses) solutions were 80 mM KCl well and 140 mM KCl pipette. Under these conditions Cl potential = -14 mV and K potential = $+14$ mV. Average reversal potential = $+16$ mV ($n = 4$, $SD = 7$), indicating cation over anion selectivity. After changing well solution to 140 mM NaCl, left I - V curve (diamonds), reversal potential shifts to negative values, indicating a slight preference for Na over K ions. Channel conductance varies with ion concentration (9). Reversal potential is obtained by linear extrapolation and thus calculated permeability ratios are estimates.

ments were carried out 2-7 days after plating. During this period cells formed monolayer islands of hundreds of cells. In no case were recordings made, either from single cells or from completely confluent culture plates.

Recording and analysis. Membranes at tips of patch pipettes allow a high temporal resolution of gating, however, they are fragile and tend to break on moving the pipette from one well to another. Planar bilayers are much more stable, enabling longer experiments but at a lower temporal resolution. The solutions used in tissue culture experiments (see Table 1) were designed to distinguish between different channels according to their

FIG. 5. An I - V relationship of 15 -pS Na channel in a planar bilayer with 150 NaCl 2 KCl *cis* and 300 NaCl 20 KCl *trans*. Reversal potential is close to Na potential (E_{Na}), demonstrating a selectivity for Na over K ($n = 9$) and over Cl ions ($n = 3$).



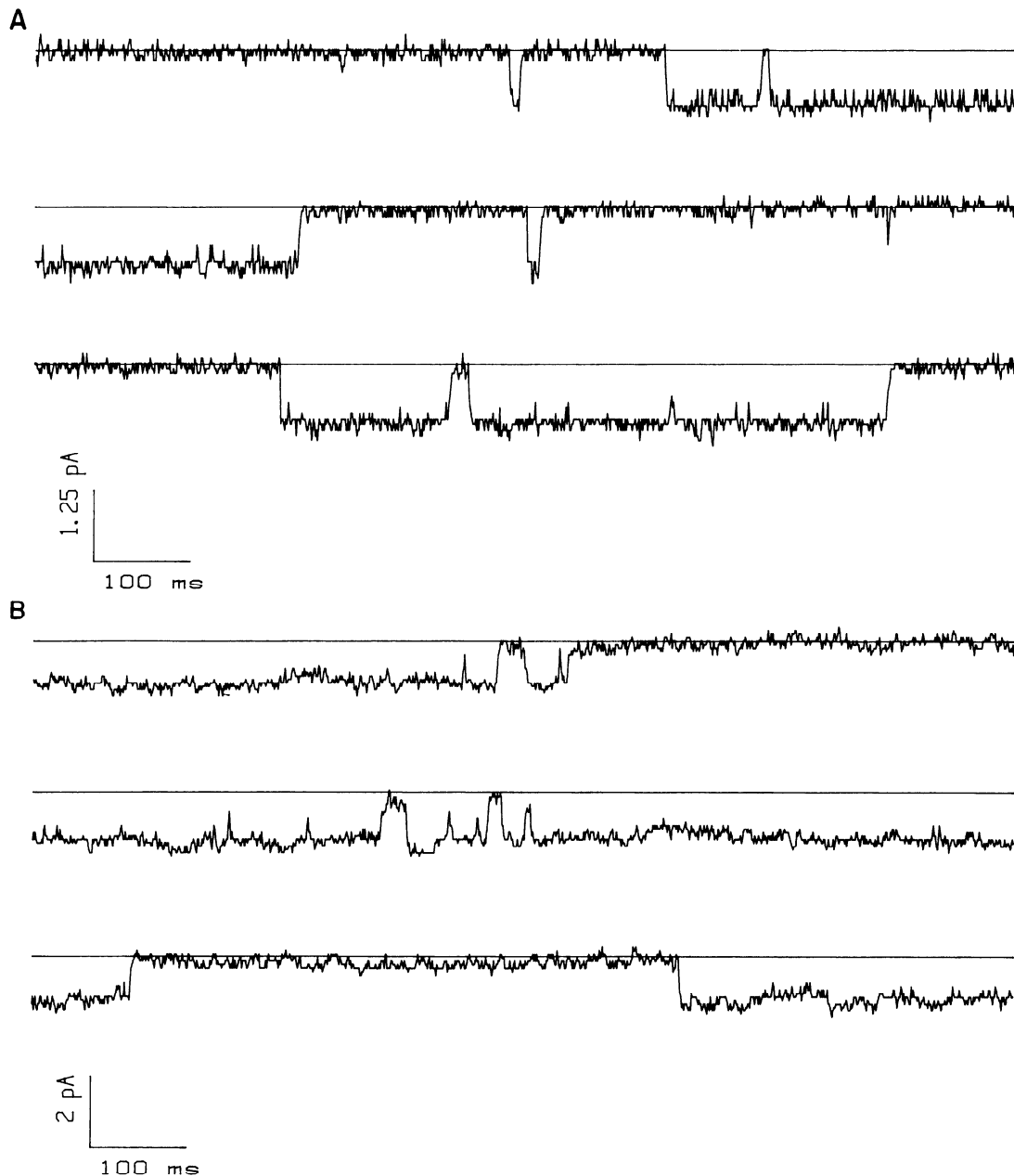


FIG. 6. Activity of a 50-pS cation-permeable channel in apical membrane of proximal convoluted tubule cells in tissue culture. A: "on cell" configuration voltage = 0 mV pipette relative to bath. B: cell-detached configuration voltage = 80 mV pipette relative to bath. Filtered at 1 kHz. For composition of solutions see Table 1.

selectivity while minimizing experimental manipulations of the patches, which are very fragile. A Yale (New Haven, CT) patch-clamp amplifier with a 10 G Ω feedback resistor was used. Data were sampled at 22 kHz (16 bit) and taped using a digital VCR recording adaptor (Medical Instrument PCM). Experiments were handled by a PDP 11/23+ computer using an INDEC laboratory display and analysis system (INDEC, Sunnyvale, CA).

RESULTS

Cation channels from BBMV. Data analyses are based on 24 records in which only a single channel was active in the membrane. In most cases channels tended to appear in clusters, even when the vesicles were diluted

to a point in which fusion events were rare, thus making individual channel analysis impossible (3).

At least two types of cation-permeable channels, differing in their kinetics and conductance were observed (Fig. 2). A 60-pS channel ($n = 15$, $SD = 13$) was observed in both tip dipping and planar bilayers. The channel current-voltage ($I-V$) curve is linear in a symmetric 300 mM NaCl solution. This channel is characterized by long open times in the range of hundreds of milliseconds. Analysis of the distribution of open- and closed-times is shown in Fig. 3A, demonstrating time constants one to two orders of magnitude apart ($n = 5$). Closed-time distributions can be fitted to two rate constants representing the bursting behaviour of this channel. The

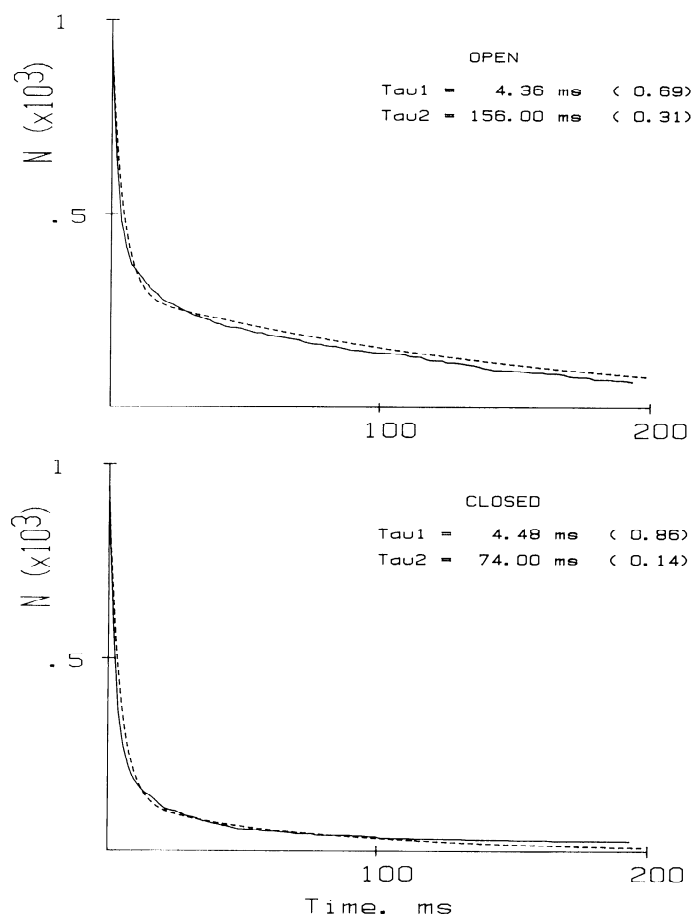


FIG. 7. Open- and closed-time distributions for 50-pS channel from cultured cells in cell-detached mode. $V = -80$ mV, bin size 5 ms; filtered at 1 kHz. Values in parentheses represent relative coefficient of each exponent in overall activity.

probability of the channel to be in the open state ranged from 0.60 to 0.95 and was independent of voltage, in the ± 100 mV range. Calcium concentrations in the range of 10^{-7} to 10^{-3} M (using EGTA as a chelator) did not affect the open-time probabilities. Measurements of permeability ratios of the channel using the Goldman-Hodgkin-Katz voltage equation (see legend of Fig. 4 for details) revealed selectivity for cations over anions [ratio of Na permeability to Cl permeability ($P_{\text{Na}}/P_{\text{Cl}}$) > 10] and a ratio of $P_{\text{K}}/P_{\text{Na}} = 1.1$ ($n = 7$, $\text{SD} = 0.23$).

A second, 15 pS (14.8 pS, $n = 9$, $\text{SD} = 1.8$) cation-selective channel, was detected in planar bilayers only. The channel I - V curve is linear in a symmetric 300 mM NaCl solution. This channel is selective to Na^+ over K^+ ($P_{\text{Na}}/P_{\text{K}} > 10$) and is not permeable to Cl^- ($P_{\text{Na}}/P_{\text{Cl}} > 10$) (Fig. 5). The channel activity is characterized by bursts and both the open- and close-time distributions can be fitted by two exponentials (Fig. 3B) with time constants one to two orders of magnitude apart ($n = 3$). The probability of the channel to be in the open state ranged from 0.4 to 0.6 independent of voltage. Calcium concentrations in the range of 10^{-7} to 10^{-3} M (using EGTA as a chelator) did not effect the open-time probability.

Cation channels in the apical membrane of PCT cells in culture. Patch-clamping the apical membrane of PCT cells in culture uncovered an assortment of channel types

in almost every patch. Among these channels were 15-pS (15.4 pS, $n = 7$, $\text{SD} = 2.2$) K^+ -selective ($P_{\text{K}}/P_{\text{Na}} = 16$, $n = 2$) calcium-activated channels, and 40-pS ($n = 3$) Cl^- channels. These and other channels are beyond the scope of the present study and are not discussed further.

A 50-pS ($n = 6$, $\text{SD} = 19$) cation-selective channel was seen in patches on apical membranes (Fig. 6). The selectivity over anions was assessed by a +50-mV reversal potential in the on cell mode. An ideal anion-selective channel would reverse at ~ 0 mV (see Table 1). When a successful transition to the detached mode was achieved, the channel current reversed between -20 and -40 mV (pipette relative to medium), indicating on the average a poor selectivity for Na^+ over K^+ ($P_{\text{Na}}/P_{\text{K}} = 1$ –5:1, mean = 2.24, $n = 5$, $\text{SD} = 1.7$). The open-time probability fluctuated both in cell-attached and inside-out configurations. The open- and close-time distributions of the channel fit a sum of two exponentials (Fig. 7) with time constants one to two orders of magnitude apart ($n = 3$). In the cell-detached mode channel gating was not affected by changing calcium concentrations in the bath from 10^{-3} to 10^{-7} M using EGTA as a calcium chelator. The channel activity is not voltage dependent.

DISCUSSION

Reconstitution of membrane protein from BBMV into artificial lipid bilayers and patch-clamping the apical side of PCT epithelial cells in culture enabled us to expose this normally inaccessible region of the nephron to single channel recordings. Using these two approaches, we demonstrate the existence of cation channels in the PCT luminal membrane of the rat. In the intermediate range, channels with a conductance of 50 pS in cell membrane patches and 60 pS in lipid bilayers, were detected by both methods. Their $P_{\text{K}}/P_{\text{Na}}$ ratio was found to be near unity. Since these channels do not discriminate between Na^+ and K^+ , under physiological conditions the major charge carrier through the channel would be Na^+ due to its much higher electrochemical gradient and resultant driving force (9).

The 15-pS channel described here is similar to the one reported by Gogelein and Greger (8) in the pars recta segment of the rabbit PCT in terms of conductance and selectivity. The channel was seen in the planar bilayer assay only and the reason for this is not clear.

Both the conductance and selectivity of the 50-pS channel recorded in tissue culture fall within the range of the values for the 60-pS channel recorded in lipid bilayer. The two may be the same or different channels. Whatever the case, one can expect that the channel gating will differ under the two experimental conditions. For reconstitution the channel protein undergoes an extensive preparatory procedure that can affect its properties. Furthermore, in the reconstituted mode the channel is devoid of all cellular modulatory mechanisms. Channel activity in primary cell cultures approximate in situ conditions more closely, nevertheless even here the system is lacking the modulation present in the intact organ.

The evidence presented here for a Na^+ conductance through cation-selective channels in the luminal mem-

brane of the PCT is consistent with several recent reports (5, 6, 8, 12). The extrapolation to the native condition remains to be substantiated by recordings in situ.

We thank E. Shurany for technical assistance.

This work was supported by US-Israel Binational Science Foundation Grant 84-00386 to D. Dagan.

Address for reprint requests: S. Marom, Dept. of Physiology and Biophysics, Faculty of Medicine, Technion-Israel Institute of Technology, POB 9649, Haifa 31096, Israel.

Received 7 October 1988; accepted in final form 20 April 1989.

REFERENCES

1. BIBER, J., B. STIEGER, W. HAASE, AND H. MURER. A high yield preparation for rat kidney brush border membranes different behaviour of lysosomal markers. *Biochim. Biophys. Acta* 647: 169–176, 1981.
2. CORRONADO, R., AND R. LATORRE. Phospholipid bilayers made from monolayers on patch clamp pipettes. *Biophys. J.* 43: 231–236, 1983.
3. COYNE, M. D., D. DAGAN, AND I. B. LEVITAN. Calcium and barium permeable channels from *Aplysia* nervous system reconstituted in lipid bilayers. *J. Membr. Biol.* 97: 205–213, 1987.
4. DARSZON, A. Planar bilayers: a powerful tool to study membrane proteins involved in ion transport. *Methods Enzymol.* 127: 486–502, 1986.
5. DRAKE-BAUMANN, R., AND E. BOULPAEP. Apical membrane channels from salamander kidney proximal tubule cells in culture (Abstract). *Biophys. J.* 53: 520a, 1988.
6. ELGAVISH, G. A., AND A. ELGAVISH. Evidence from ^{23}Na NMR studies for the existence of sodium channels in the brush border membrane of the renal proximal tubule. *Biochem. Biophys. Res. Commun.* 128: 746–753, 1985.
7. EVERS, C., W. HASSE, H. MURER, AND R. KINNE. Properties of brush border vesicles isolated from rat kidney cortex by calcium precipitation. *Membr. Biochem.* 1: 203–219, 1978.
8. GOGELIN, H., AND R. GREGER. Na selective channels in the apical membrane of rabbit late proximal tubules. *Pfluegers Arch.* 406: 198–203, 1986.
9. HILLE, B. *Ionic Channels of Excitable Membranes*. Boston, MA: Sinauer, 1984.
10. LANG, F., G. MESSNER, AND W. REHWALD. Electrophysiology of sodium-coupled transport in proximal renal tubules. *Am. J. Physiol.* 250 (Renal Fluid Electrolyte Physiol. 19): F953–F962, 1986.
11. MARKWELL, M. A. K., S. M. HAAS, L. L. BIEBER, AND N. E. TOLBERT. A modification of the Lowry procedure to simplify protein determination in membrane and lipoprotein samples. *Anal. Biochem.* 87: 206–210, 1978.
12. MEROT, J., M. BIDET, B. GACHOT, S. LE MAOUT, M. GAUC, AND P. POUJEOL. Patch-clamp study on primary culture of isolated proximal convoluted tubules. *Pfluegers Arch.* 413: 51–61, 1988.
13. VINAY, P., A. GOUGOUX, AND G. LEMIEUX. Isolation of a pure suspension of rat proximal tubules. *Am. J. Physiol.* 241 (Renal Fluid Electrolyte Physiol. 10): F403–F411, 1981.
14. WEINSTEIN, A. M., AND E. E. WINDHAGER. Sodium transport along the proximal tubule. In: *The Kidney: Physiology and Pathophysiology*, edited by D. W. Seldin and G. Giebisch. New York: Raven, 1985, p. 1033–1061.

